

at the required moisture level and set this voltage in the Arduino™ sketch. Measure the voltage between analog pin 0 and ground of the Arduino™ board in the above circuit to calibrate the system to the required moisture level.

3. Then use the Digital Pin 13 to turn on the motor if the soil is dry.

To switch a motor ON or OFF, we need to use a relay. However, the relay itself needs very high current. It, therefore, needs to be connected using the circuit described with the given Transistor 2N2222.

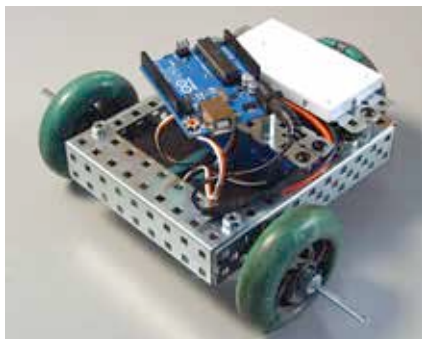
Check more details about the project at <http://dgit.in/garduinoO>

Wireless computer-controlled toy car

A wireless toy car is a basic version of advanced and equipped vehicles used for surveillance.

To make a toy car, you'll need the following components:

- ◆ Motors
- ◆ Wheels
- ◆ Chassis
- ◆ RF transmitter and receiver
- ◆ Battery
- ◆ Motor shield



Wireless toy car

How to implement:

You'll need to use the Motor shield along with the Arduino™ board to drive the motors connected to the toy car. (The code for this is explained in Chapter 6.)

To steer the 4-wheeled vehicle, you'll use the driving mechanism called 'differential' drive. It uses the following motor rotation directions to steer the toy car:

	Right Wheel - Forward Rotation	Right Wheel - Forward Backwards Rotation
Left Wheel - Forward Rotation	Vehicle moves forward (w)	Vehicle turns right (d)
Left Wheel - Backwards Rotation	Vehicle turns left (a)	Vehicle moves backwards (s)